

Program -7

Deploy a Web Application to Kubernetes using Minikube

1. Project Structure

```
Prgm7/  
|  
|--- nginx-deployment.yaml
```

2. Configuration files

Create **nginx-deployment.yaml**

```
apiVersion: apps/v1  
kind: Deployment  
metadata:  
  name: nginx-deployment  
spec:  
  replicas: 2  
  selector:  
    matchLabels:  
      app: nginx  
  template:  
    metadata:  
      labels:  
        app: nginx  
    spec:  
      containers:  
        - name: nginx  
          image: nginx:latest  
          imagePullPolicy: IfNotPresent  
          ports:  
            - containerPort: 80
```

3. Start Minikube

Minikube start

4. Check the status of Minikube

Minikube status

5. Apply the nginx-deployment.yaml file

docker apply -f nginx-deployment.yaml

6. Check status

docker get pods

docker get deployments

7. Expose the deployment as service

kubectl expose deployment nginx-deployment - --type=NodePort - --port=80

8. Check services

kubectl get services

9. Access the web app

minikube service nginx-deployment

Output:-

```
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn7$ minikube service nginx-deployment
Type: NodePort
IP Family Policy: SingleStack
IP Families: IPv4
IP: 10.111.134.55
IPs: 10.111.134.55
Port: <unset> 80/TCP
TargetPort: 80/TCP
NodePort: <unset> 31128/TCP
Endpoints: 10.244.0.119:80,10.244.0.120:80
Session Affinity: None
External Traffic Policy: Cluster
Internal Traffic Policy: Cluster
Events: <none>
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn7$
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:31128

```
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn7$
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org. Commercial support is available at nginx.com.

Thank you for using nginx.